

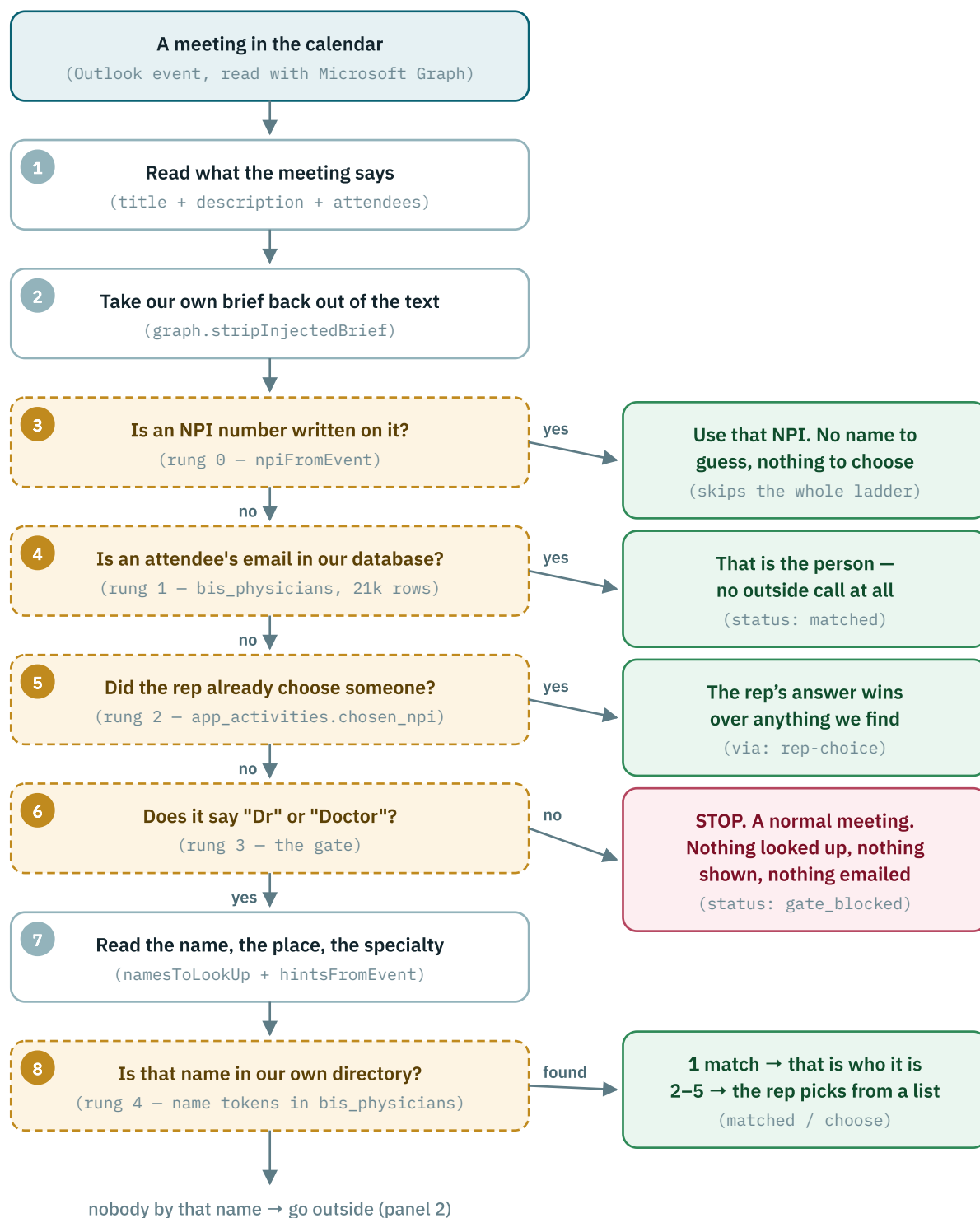
How the agent finds the physician

The whole path from a meeting in the calendar to a pre-meeting brief — in plain English, with the name used in the code in brackets. Every step below is a real step in `src/meeting-match.js`, `src/outside-sources/*` and `src/enrichment/*`.

Outlook Calendar Intelligence · pre-meeting sales intelligence · 3
September 2026

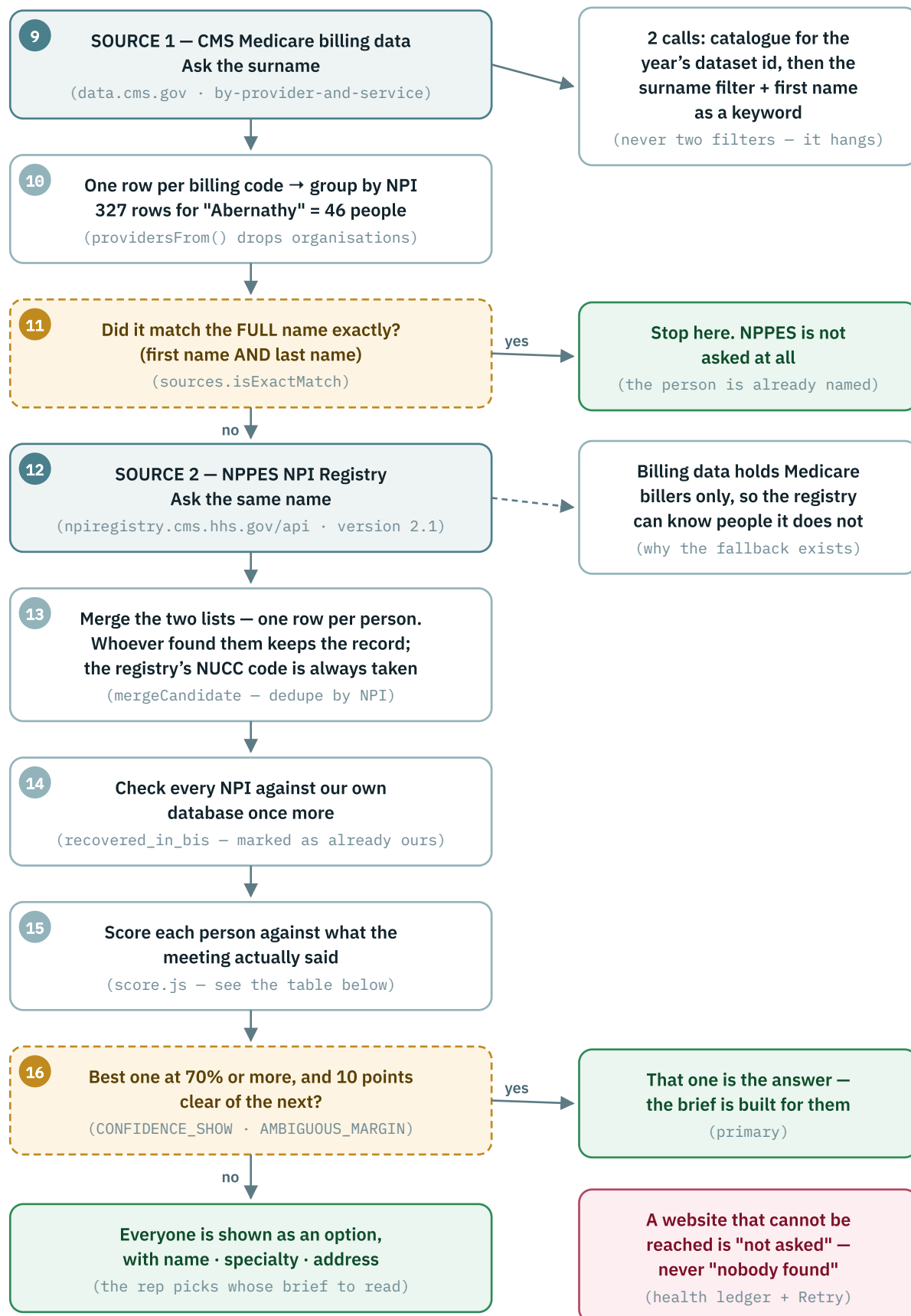
1 · What we already know

The cheapest answers first: our own data, and the rep's own decision



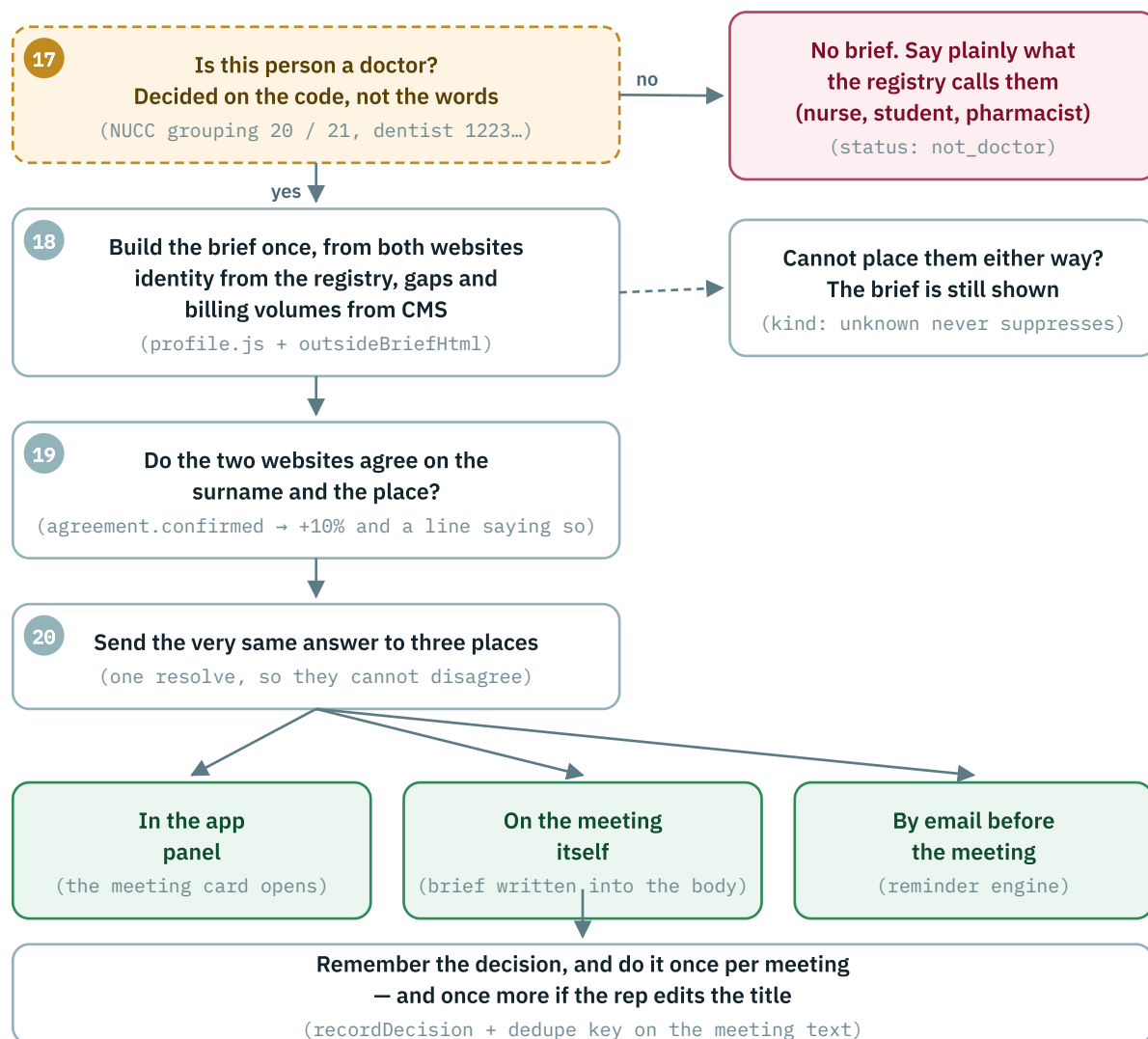
2 · Outside our database — the two public websites

Billing data first, the registry second; both are plain JSON APIs, no scraping



3 · Is it a doctor? → the brief → where it goes

A brief is produced for physicians, dentists and podiatrists; everybody else is named, not briefed



Where the data actually comes from

Both websites are read through their own public JSON APIs. Nothing is scraped, and no key is needed. What a person sees on the website is what the API returns.

1 · NPPES NPI Registry — turns a name into an NPI

```
GET https://npiregistry.cms.hhs.gov/api/
  ?version=2.1
  &enumeration_type=NPI-1          # a person; NPI-2 would be a hospital
  &last_name=ABERNATHY&first_name=JOHN
  &state=FL&limit=20
```

WHAT WE TAKE FROM IT	WHY IT MATTERS
NPI, name, credential	the NPI is this app's primary key everywhere
Primary taxonomy + its NUCC code	the code decides whether a brief is produced at all
Practice address, city, state, ZIP, phone	what a rep needs to recognise the right person
Licence number and state, NPI date, status	shown as <i>extra</i> — our own database has no column for these
No email, ever	the registry has no email field, so an email is never guessed from it

Three real quirks, handled in the code. A one-letter wildcard is rejected, so a single initial is never sent — it is kept as a ranking hint. A bad query comes back as HTTP 200 with an `Errors` array, not as an error. And the same query can return 0 results once and 1 a moment later, so one extra attempt is made (`retryIfEmpty`).

2 · CMS Medicare Physician & Other Practitioners — by provider and service

```
# step 1 — which dataset is this data year? (ids change when CMS republishes)
GET https://data.cms.gov/data.json          → 2024 → 92396110-2aed-4d63-a6a2-5d6207d46a29

# step 2 — ask the surname (and the first name as a keyword)
GET https://data.cms.gov/data-api/v1/dataset/<uuid>/data
  ?filter[Rndrng_Privr_Last_Org_Name]=Abernathy
  &keyword=John
  &column=Rndrng_NPI,Rndrng_Privr_First_Name,Rndrng_Privr_Type,...
  &size=500

# step 3 — later, for the chosen person: their billing lines, year by year
GET .../data?filter[Rndrng_NPI]=1265847438&size=500
```

MEASURED BEHAVIOUR	CONSEQUENCE IN THE CODE
Surname filter: 327 rows in ~0.5 s	one row per <i>billing code</i> , so rows are grouped by NPI — 327 rows = 46 people
Surname filter + keyword: precise and fast	the first name is a keyword, never a second filter
Two <code>filter[...]</code> params: the request hangs	never send two — this is a hard rule
<code>Rndrng_Privr_Ent_Cd = I / O</code>	organisations are dropped — a clinic is not who the rep is meeting
Provider type in CMS's own words, no NUCC code	"Family Practice", "Podiatry" — read by word, and the registry's code wins when both exist

How a person is scored against the meeting

Every candidate starts at 30 for being in a registry at all. Then the meeting's own words are checked *for* the candidate's values — asking "does this meeting mention CHICAGO?" rather than trying to guess a city out of free text.

EVIDENCE FROM THE MEETING	POINTS
In a registry at all (the starting point)	+30
Last name matches	+25
First name matches exactly	+30
First name partial / initial only	+15 / +10
First name is a different name	-25
State matches / city matches	+15 / +15
The meeting mentions this specialty (taxonomy)	+20
Specialty matches exactly / differs	+25 / -15
ZIP, street address or phone matches	+20 each
Only candidate anybody returned	+10
Both websites confirm the same identity	+10

Two thresholds, and they are the rep's own rule. 70% and at least 10 points clear of the next candidate → the data is put in front of the rep as the answer. Anything less → everybody is listed as an option, with name, specialty and practice address, and the rep picks. A shortlist is never hidden just because nothing cleared the bar: "here are the five people called Ajjarapu" is an answer a rep can act on.

The same flow, on three real meetings

THE MEETING SAYS	WHO IS ASKED	WHAT THE REP SEES
Meeting with Dr JOHN ABERNATHY	CMS only — it matched the full name, so the registry is not asked	One person: NPI 1265847438, Internal Medicine, St Petersburg FL → the brief opens
Meeting with Dr Abernathy	CMS (46 people) and then the registry — a surname is never an exact match	A list, each row with specialty and address, all at 55% — the rep picks; a nurse practitioner and an optometrist are named but not offered
Meeting with Dr Avanthi Ajjarapu	CMS has nobody (she bills no Medicare) → the registry answers exactly	One person: Obstetrics & Gynecology, Fort Worth TX → the brief opens
Meeting with Best friend	Nobody. The meeting never says "Dr"	Nothing at all — a plain calendar row that does not even open

Generated by `scripts/agent-flow-doc.py`. The behaviour drawn here is covered by the test suite (`npm test`, 169 tests) and was verified against the live APIs on 2 September 2026. Numbers in circles match the step numbers used in the walk-through.